



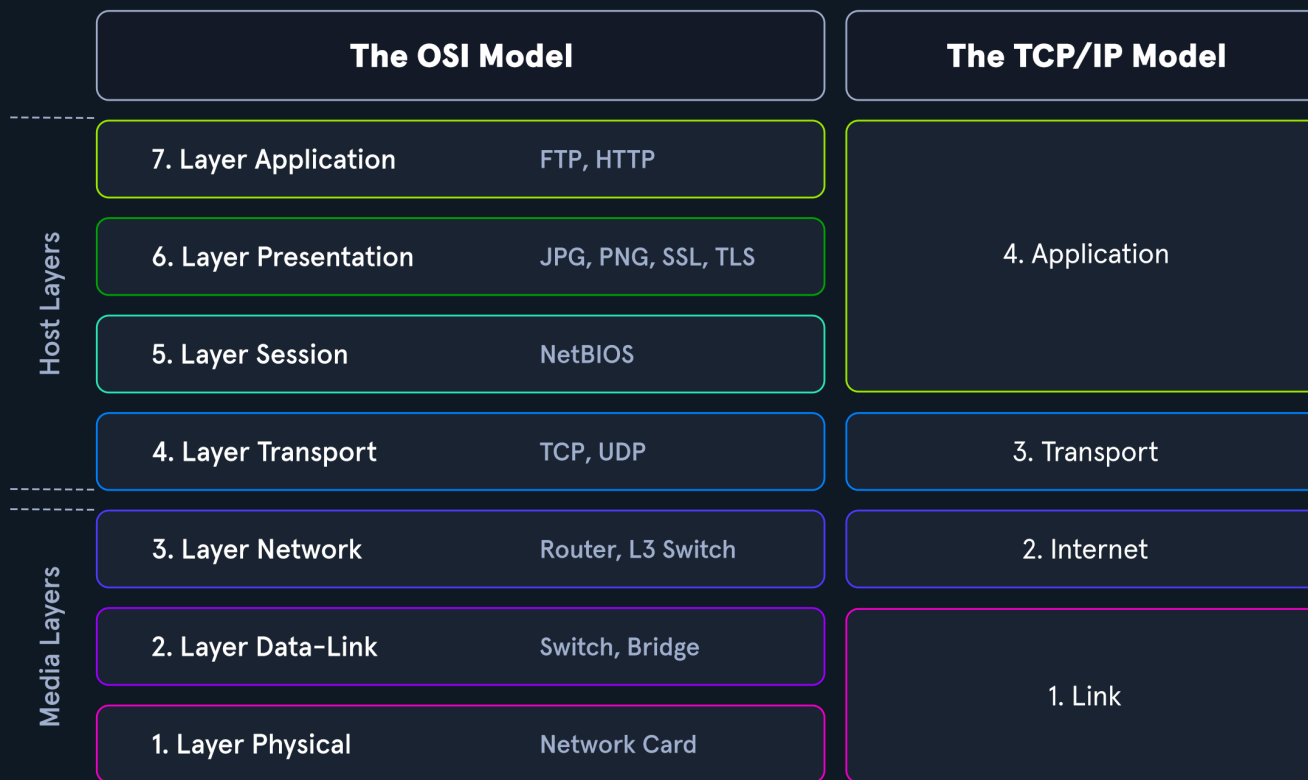
The TCP/IP Model

The **TCP/IP** model is also a layered reference model, often referred to as the **Internet Protocol Suite**. The term **TCP/IP** stands for the two protocols **Transmission Control Protocol (TCP)** and **Internet Protocol (IP)**. **IP** is located within the **network layer (Layer 3)** and **TCP** is located within the **transport layer (Layer 4)** of the **OSI** layer model.

Layer	Function
4.Application	The Application Layer allows applications to access the other layers' services and defines the protocols applications use to exchange data.
3.Transport	The Transport Layer is responsible for providing (TCP) session and (UDP) datagram services for the Application Layer.
2.Internet	The Internet Layer is responsible for host addressing, packaging, and routing functions.

Layer	Function
1.Link	The Link layer is responsible for placing the TCP/IP packets on the network medium and receiving corresponding packets from the network medium. TCP/IP is designed to work independently of the network access method, frame format, and medium.

With **TCP/IP**, every application can transfer and exchange data over any network, and it does not matter where the receiver is located. **IP** ensures that the data packet reaches its destination, and **TCP** controls the data transfer and ensures the connection between data stream and application. The main difference between **TCP/IP** and **OSI** is the number of layers, some of which have been combined.



The most important tasks of **TCP/IP** are:

Task	Protocol	Description
Logical Addressing	IP	Due to many hosts in different networks, there is a need to structure the network topology and logical addressing. Within TCP/IP, IP takes over the logical addressing of networks and nodes. Data packets only reach the network where they are supposed to be. The methods to do so are <code>network classes</code> , <code>subnetting</code> , and <code>CIDR</code> .
Routing	IP	For each data packet, the next node is determined in each node on the way from the sender to the receiver. This way, a data packet is routed to its receiver, even if its location is unknown to the sender.
Error & Control Flow	TCP	The sender and receiver are frequently in touch with each other via a virtual connection. Therefore control messages are sent continuously to check if the connection is still established.
Application Support	TCP	TCP and UDP ports form a software abstraction to distinguish specific applications and their communication links.
Name Resolution	DNS	DNS provides name resolution through Fully Qualified Domain Names (FQDN) in IP addresses, enabling us to reach the desired host with the specified name on the internet.



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